



SAIR

Spatial AI & Robotics Lab

CSE 473/573

L5: MORPHOLOGY

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Morphology – Introduction

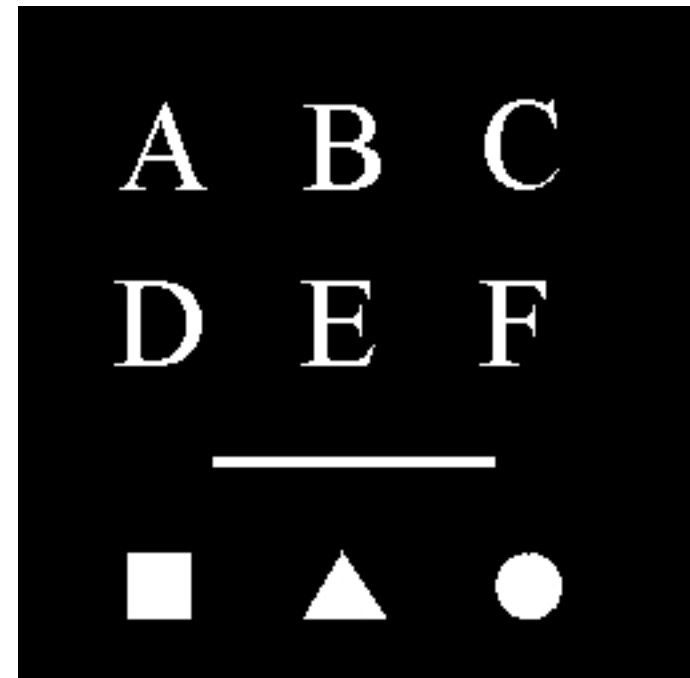
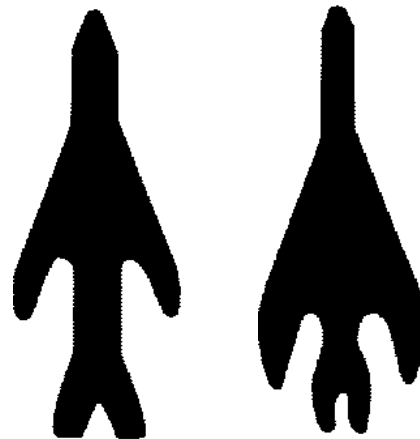
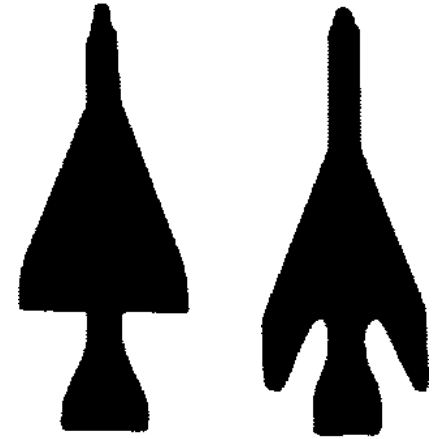
Looking at these images.....

What is interesting, important or useful information we care about?

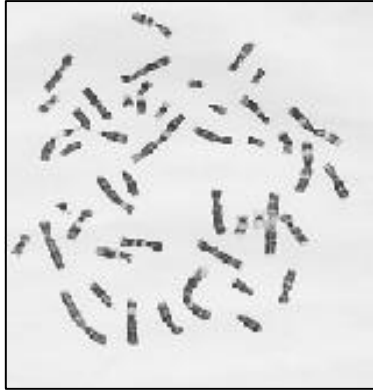
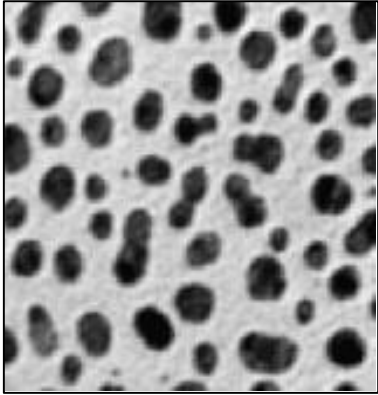
The **pixel value** of the image is **not important** as there are only **two** different values.

➤ Region shape and boundaries of object are **important**.

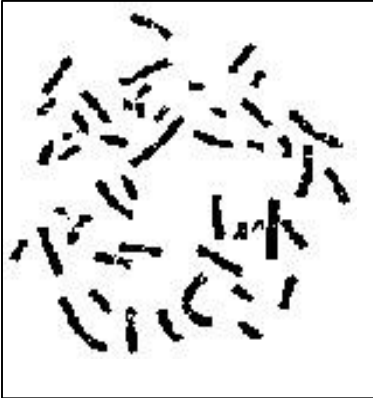
➤ Form and structure can be represented by object **pixel set**.



Morphology – Introduction



Grayscale Images



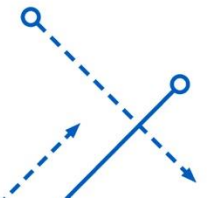
Binary Images

Image analysis needs to measure the **characteristics of objects** in the images.

Geometric measurements are important objects characteristics

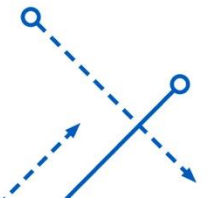
- location, orientation, area, length of perimeter

These geometric characteristics are often easier to be measured from **binary images**.



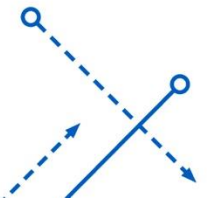
Morphology – Introduction

- Visual perception requires image processing to extract **shape information**.
- **Goal:**
 - Distinguish meaningful shape information from irrelevant one.
- The vast majority of shape processing and analysis techniques are **based on designing a shape operator** which satisfies desirable properties.



Morphology – Introduction

- **Morphology** deals with **form and structure**
- Mathematical morphology is a tool for **extracting image components** useful in:
 - representation and description of region **shape** (e.g. **boundaries**)
 - pre- or post-processing (filtering, thinning, etc.)
- Morphological operations usually follow a segmentation task or an edge detection task.
 - Thus, **often** operate on **binary images**.
- Based on **set theory** and **logic operations**



Morphology – Set Theory

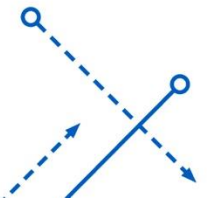
Know the
Terminology

- A two-dimensional integer space is denoted by $\mathbf{Z}^2$.
- An **element** in this space has two components $a=(a_1, a_2)$.
- For image representation, $a=(a_1, a_2)$ are the x - and y -coordinates of a pixel.
- Let A be a **set** in $\mathbf{Z}^2$. If $a=(a_1, a_2)$ is an element of A , we denote

$$a \in A$$

- If not, then

$$a \notin A$$



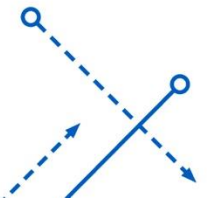
Morphology – Set Theory

- $\emptyset$ denotes null (empty) set
- An example that specifies a set C :

$$C = \{ w \mid w = a + d, a \in A \}, d = (8, 5).$$

- If a set A is a **subset** of B , we denote:

$$A \subseteq B$$



Morphology – Set Theory

- Union of A and B :

$$C = A \cup B$$

- Intersection of A and B :

$$D = A \cap B$$

- Disjoint sets:

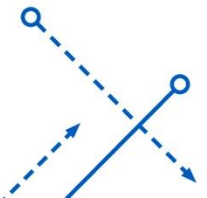
$$A \cap B = \emptyset$$

- Complement of A :

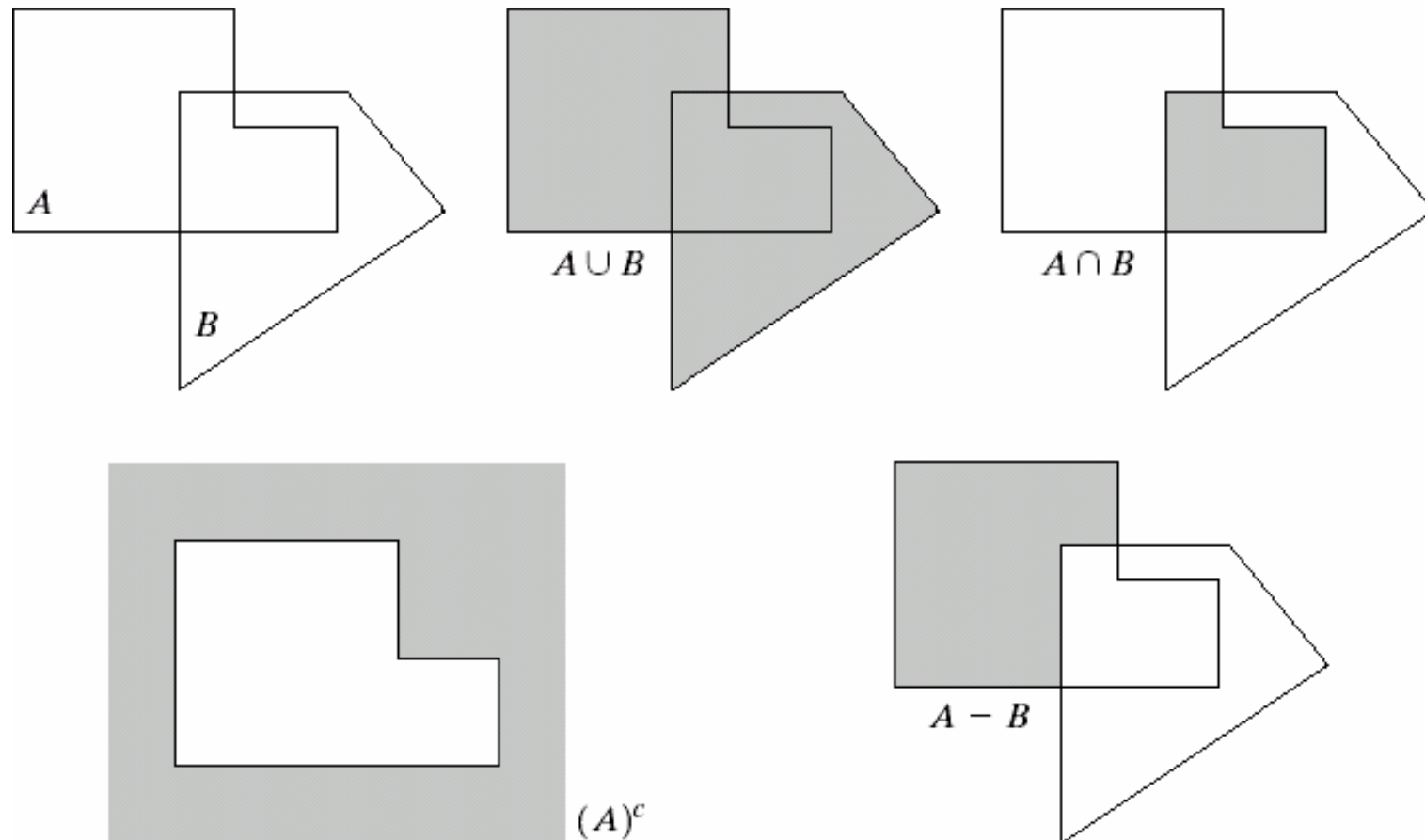
$$A^c = \{w \mid w \notin A\}$$

- Difference of A and B :

$$\begin{aligned} A - B &= \{w \mid w \in A, w \notin B\} \\ &= A \cap B^c \end{aligned}$$



Morphology – Set Theory

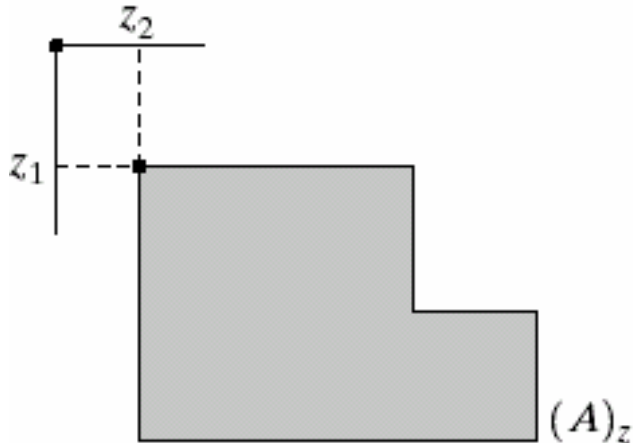


a	b	c
d	e	

(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .

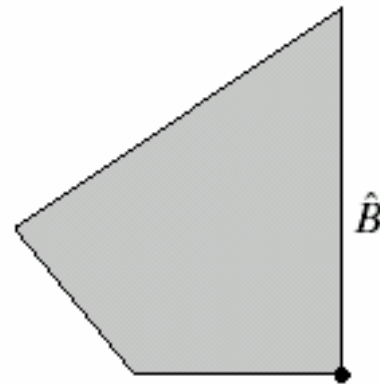
Morphology – Set Theory

- Translation of A by $z=(z_1, z_2)$: $(A)_z = \{c \mid c = a+z, a \in A\}$



a b

- (a) Translation of A by z .
- (b) Reflection of B .



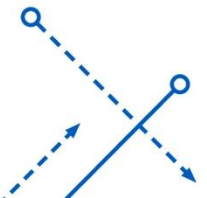
- Reflection of B :

$$\hat{B} = \{w \mid w = -b, b \in B\}$$

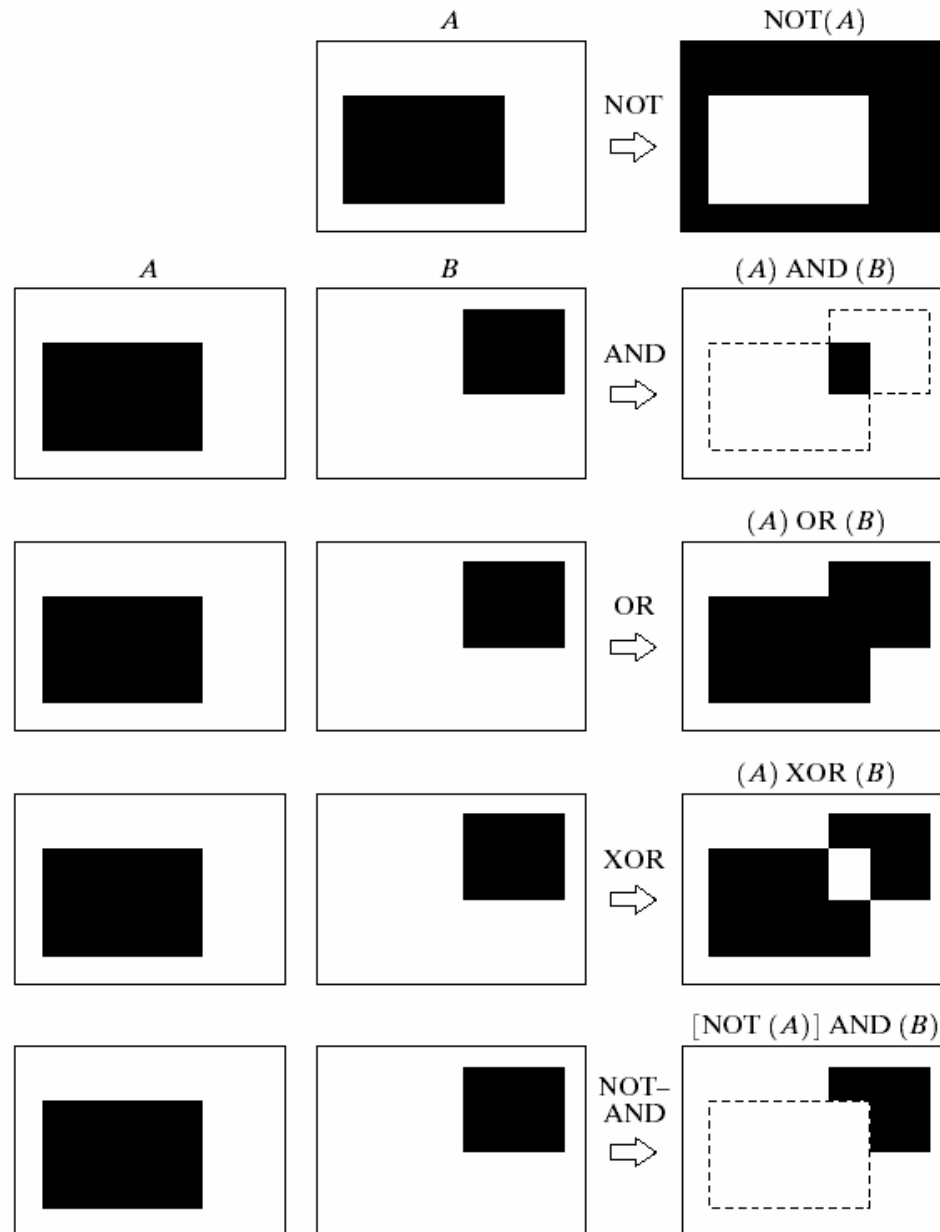
Morphology – Set Theory

- Three basic logical operations

p	q	p AND q (also $p \cdot q$)	p OR q (also $p + q$)	NOT (p) (also $\bar{p}$)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

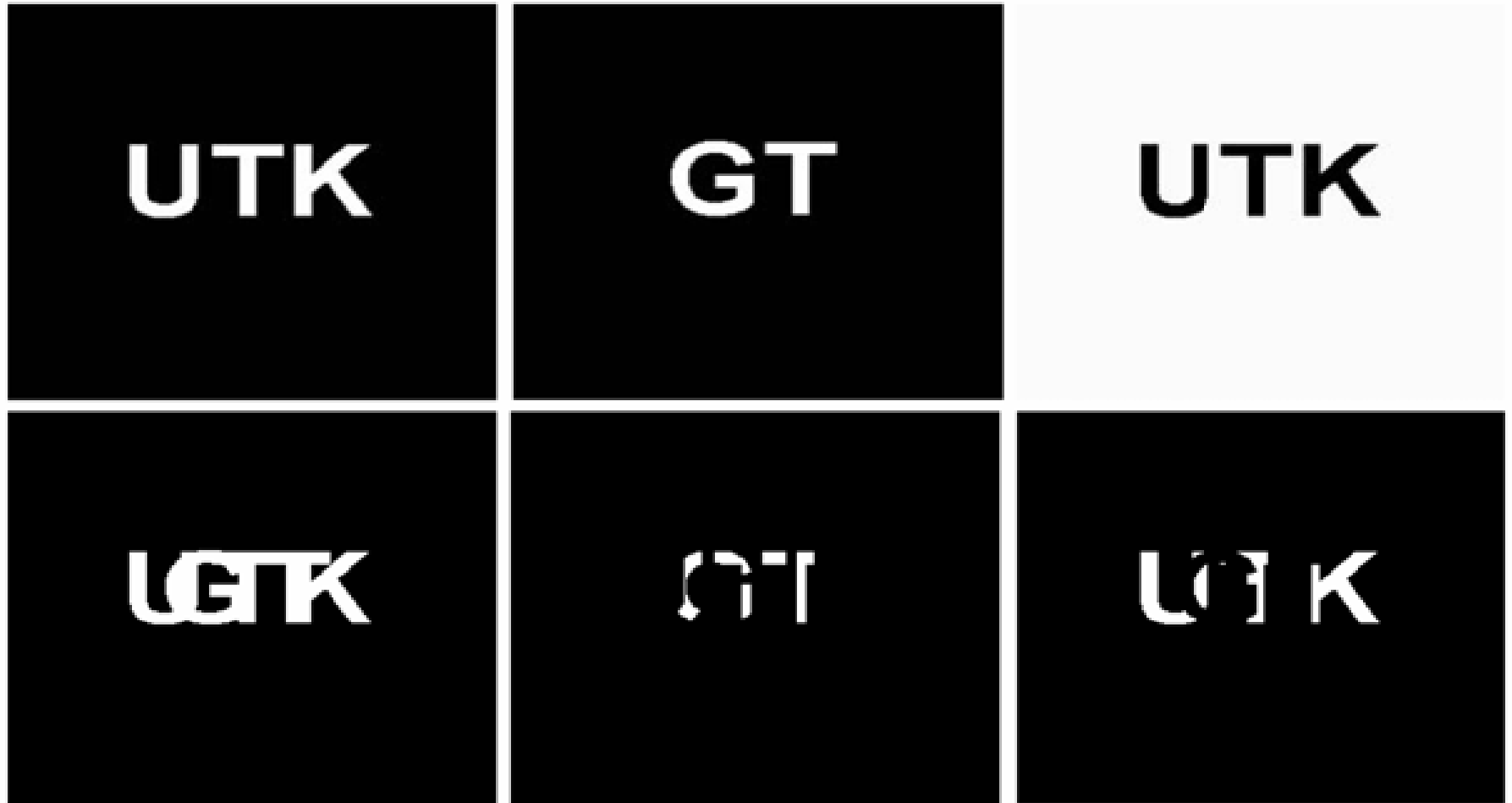


Morphology



Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Morphology

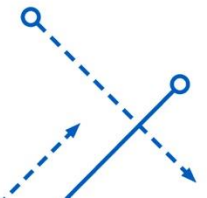


a	b	c
d	e	f

(a) Binary image A. (b) Binary image B. (c) Complement $\sim A$. (d) Union $A \cup B$. (e) Intersection $A \cap B$. (f) Set difference $A - B$

Morphology – Operators

- Primary morphological operations are **Dilation** and **Erosion**
- More complicated morphological operators such as **Opening** and **Closing** can be designed by combining erosions and dilations
- **Opening** generally **smoothes the contour** of an image and **eliminates protrusions**
- **Closing** **smoothes sections of contours**, but it generally **fuses** breaks, holes and gaps



Morphology – Dilation

Why
Reflection of
B?

- **Dilation** of A by B , denoted by $A \oplus B$, is defined as:

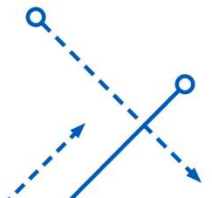
$$A \oplus B = \{z \mid \left[\left(\hat{B} \right)_z \cap A \right] \neq \emptyset \}$$

- **Interpretation:**

Obtaining the reflection of B about its origin and then shifting by z .

Dilation of A by B is the set of all z displacements such that $\hat{B}$ and A overlap by at least one nonzero element.

- B is called the **structuring element** in Dilation.



Morphology – Dilation

- **Dilation** of A by B can also be expressed as:

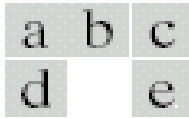
$$A \oplus B = \{ z \mid \left[\left(\hat{B} \right)_z \cap A \right] \subseteq A \}$$

- **Further Interpretation:**

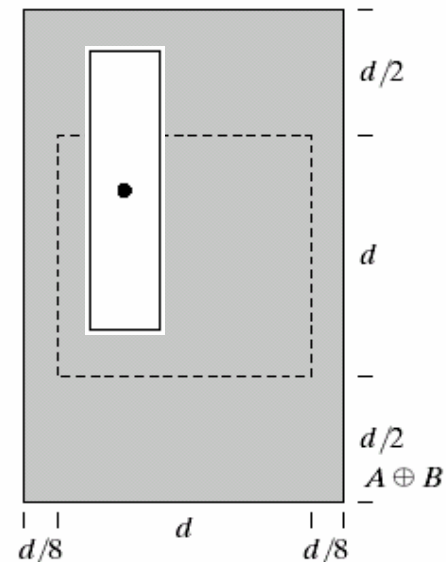
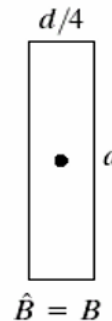
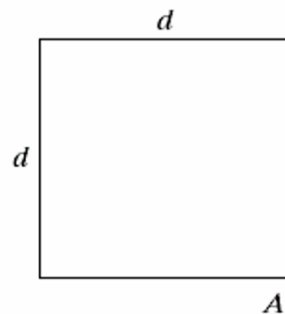
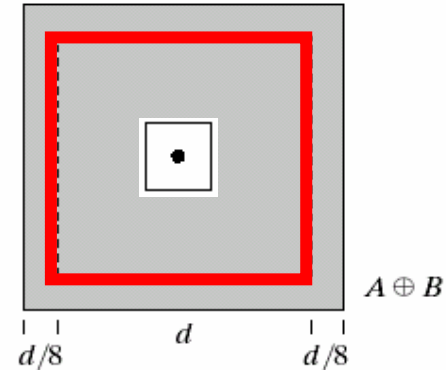
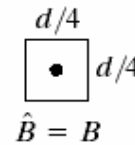
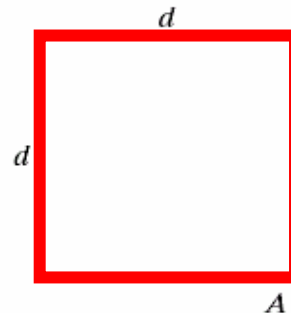
Set B can be viewed as a convolution mask.

The process of “flipping” B and then successively displace it so that it slides over set (image) A is analogous to the convolution.

Morphology – Dilation



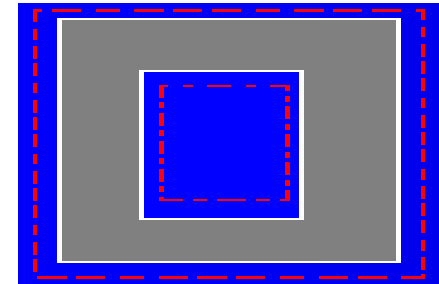
- (a) Set A .
- (b) Square structuring element (dot is the center).
- (c) Dilation of A by B , shown shaded.
- (d) Elongated structuring element.
- (e) Dilation of A using this element.



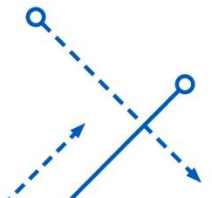
Morphology – Dilation

- The dilation morphological operation generates an output image g from an input image f using a structuring element h :

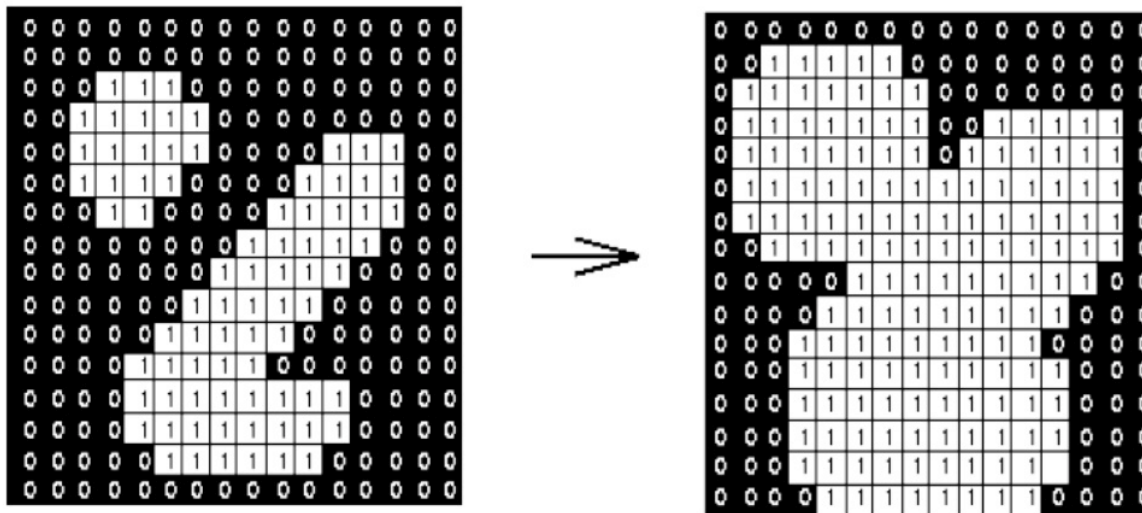
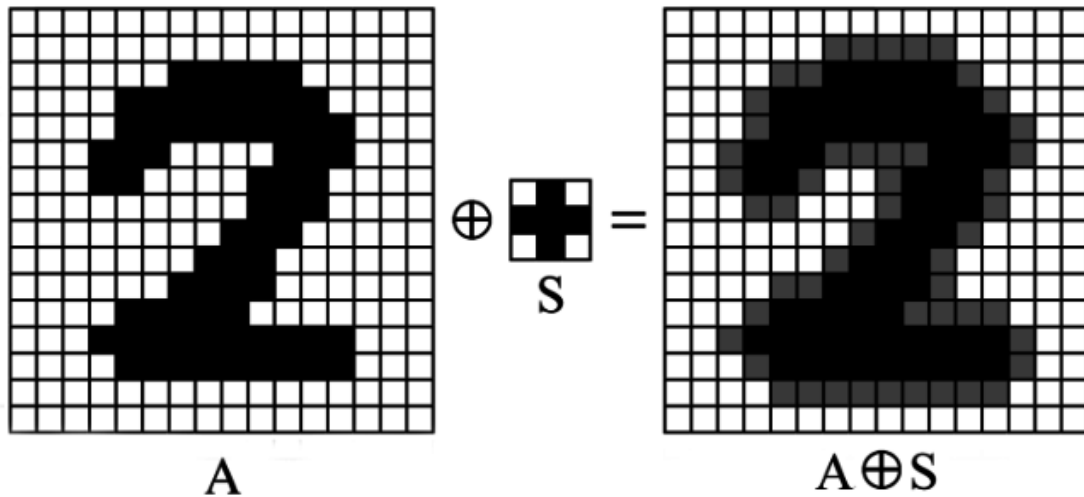
$$g(x, y) = \begin{cases} 1, & \text{if } h \text{ hints } f \\ 0, & \text{else} \end{cases}$$



- The effect of dilation with 3×3 mask is to add a single layer of pixels to the outer edge of an object and to decrease by a single layer of pixels to the holes in the object.
- A 5×5 mask will add two layers of pixels which is equivalent to applying a 3×3 mask twice.
- The main application of dilation is to remove small holes from the interior of an object.



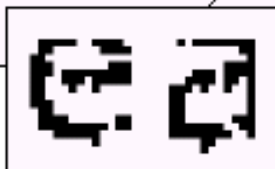
Dilation - Example



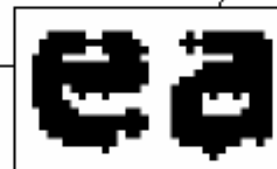
Effect of dilation using a 3×3 square structuring element

Dilation - Application

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



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0	1	0
1	1	1
0	1	0

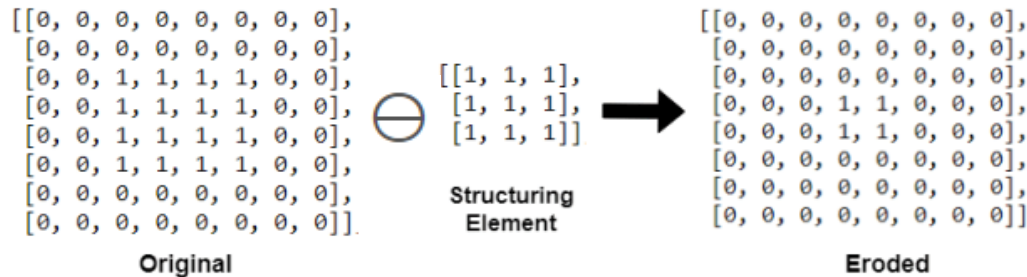
a c
b

(a) Sample text of poor resolution with broken characters (magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

Morphology - Erosion

- Erosion of A by B , denoted $A \ominus B$, is defined as:

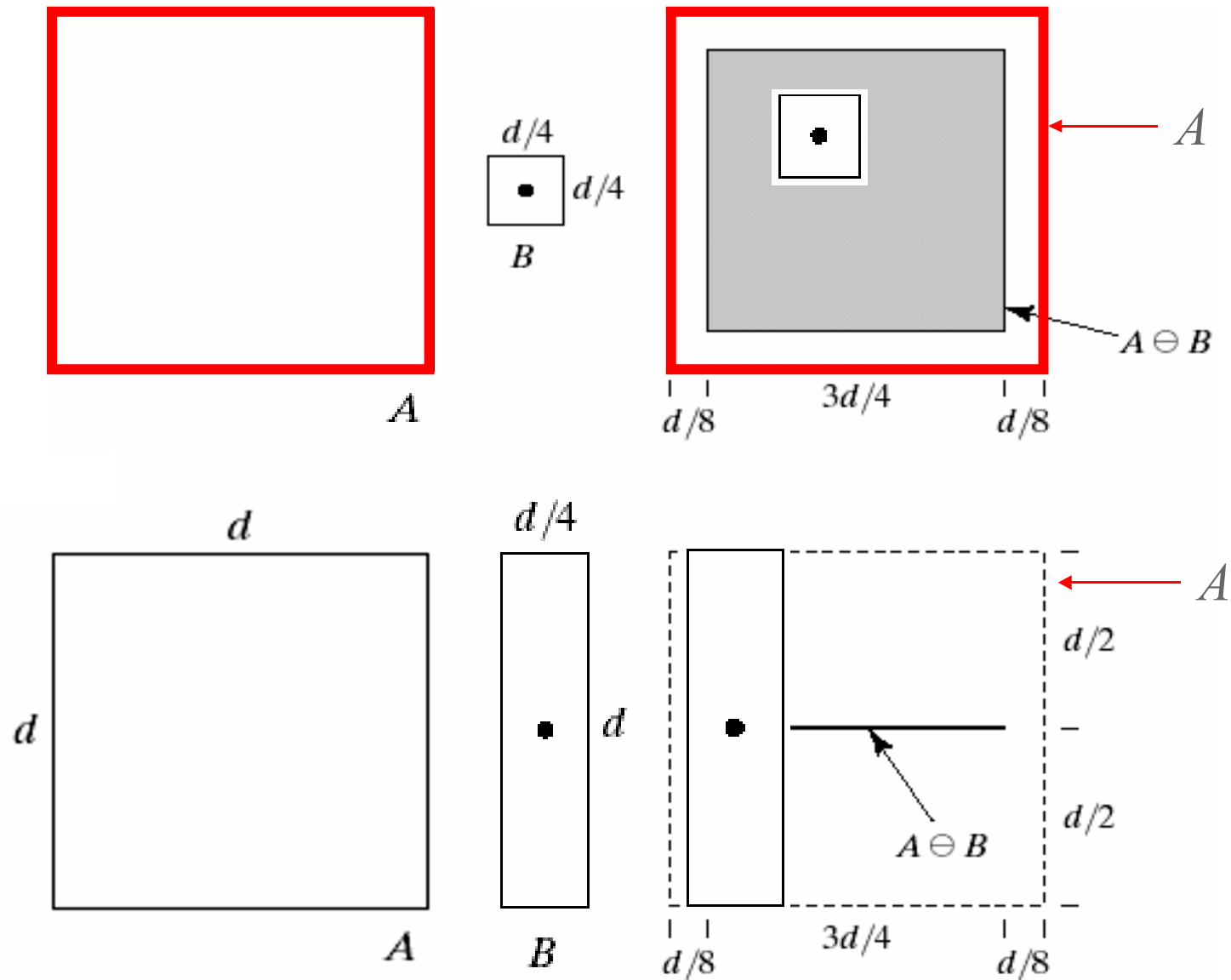
$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$



- Erosion of A by B is the set of all points z such that B , translated by z , is contained in A .
- Dilation and erosion are duals of each other with respect to set complementation and reflection. That is,

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

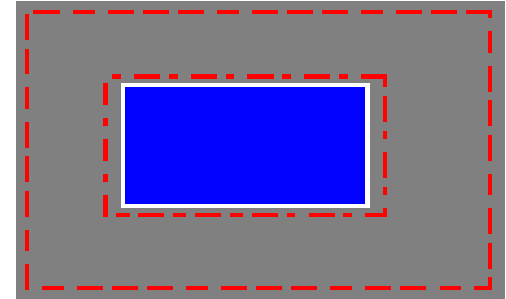
Erosion - Example



Erosion

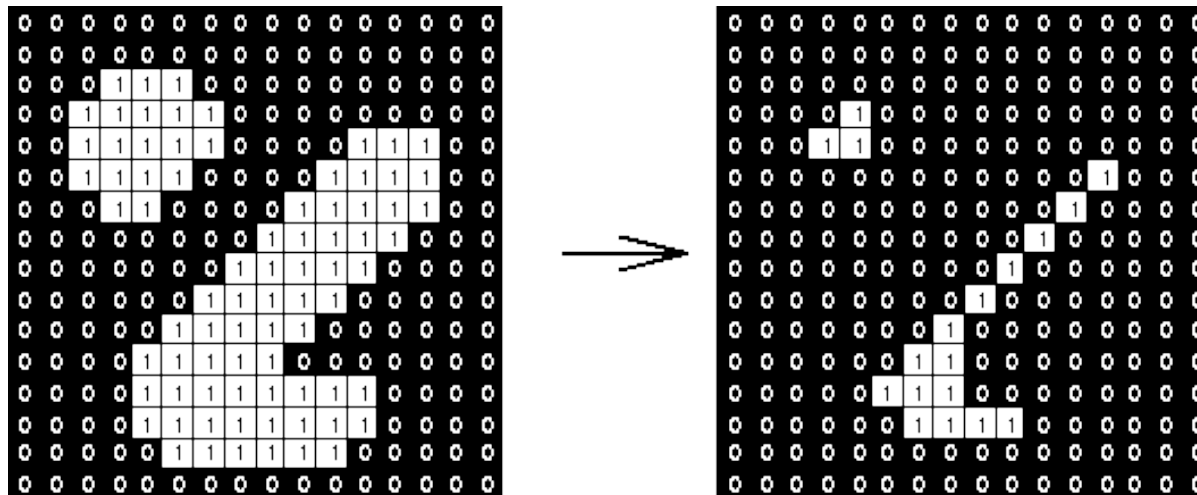
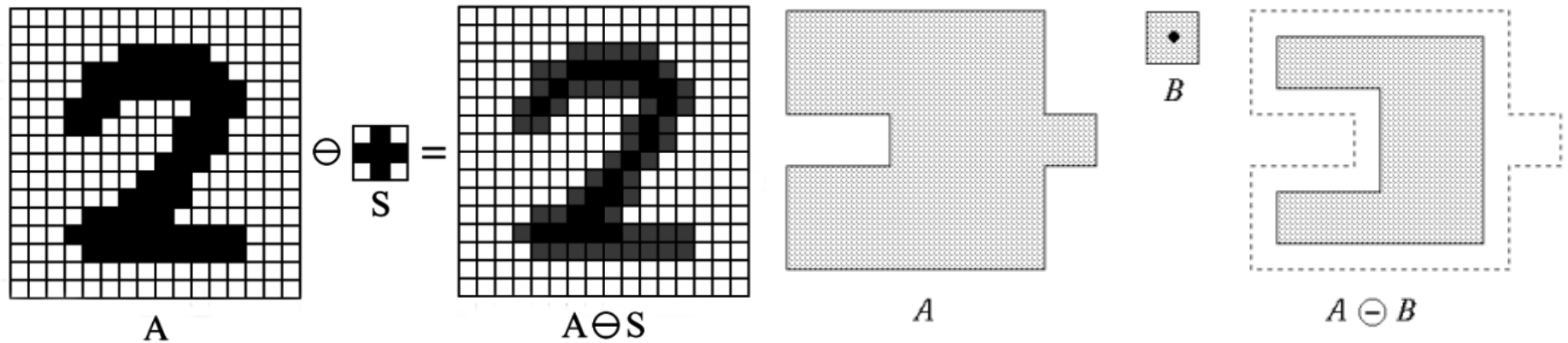
- The erosion operation generates an output g from an input f using a structuring element h where :

$$g(x, y) = \begin{cases} 1, & \text{if } h \text{ completely falls in } f \\ 0, & \text{else} \end{cases}$$



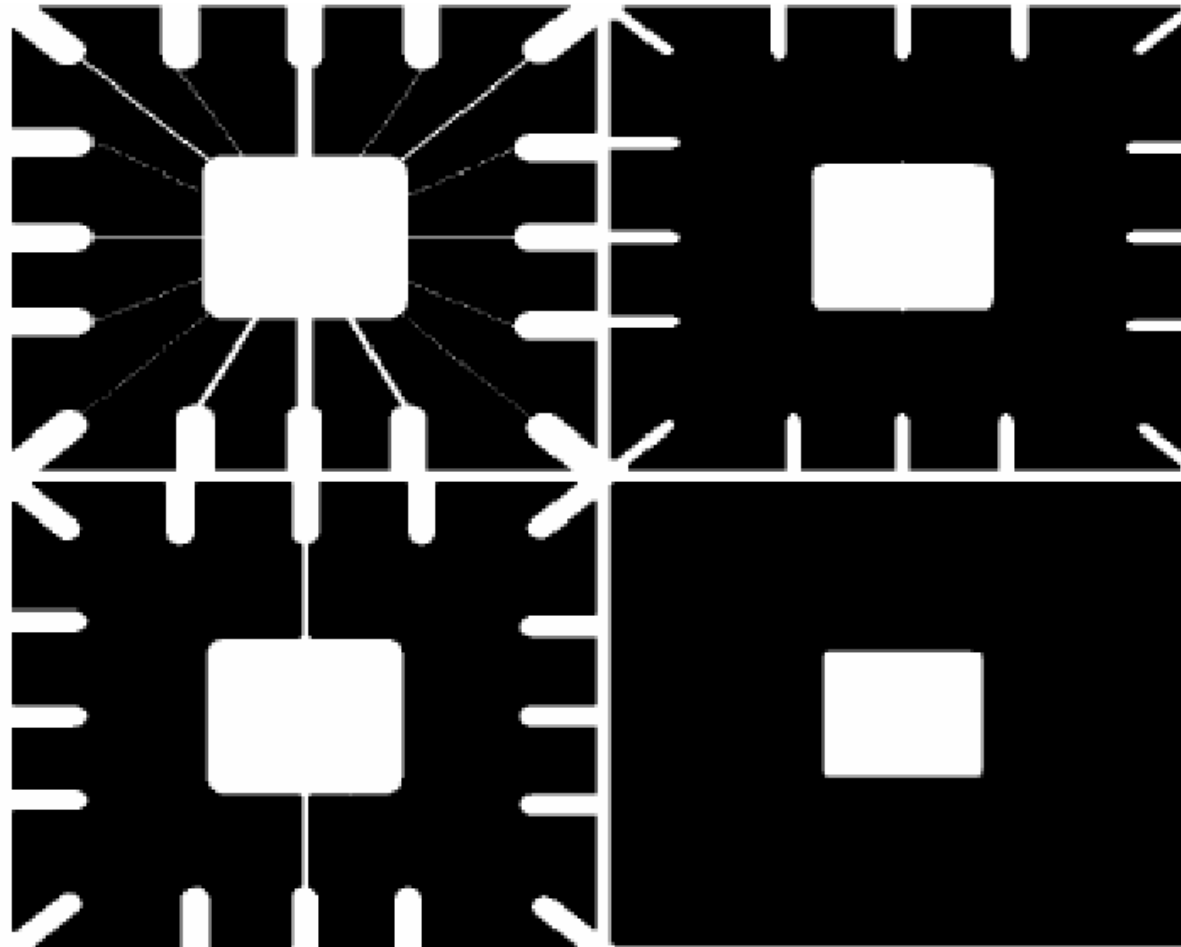
- The effect of an erosion with 3×3 mask is to strip a **single** layer of pixels from the **outer edge** of an object and to **increase** by a **single layer** of pixels to **holes** in the object.
- A 5×5 mask will strip off **two** layers of pixels which is equivalent to applying a 3×3 mask twice.
- The main application of erosion is to **remove small noise artifacts** from an image.

Erosion - Example



Effect of erosion using a 3x3 square structuring element

Morphology – Erosion



a b
c d

An illustration of erosion.

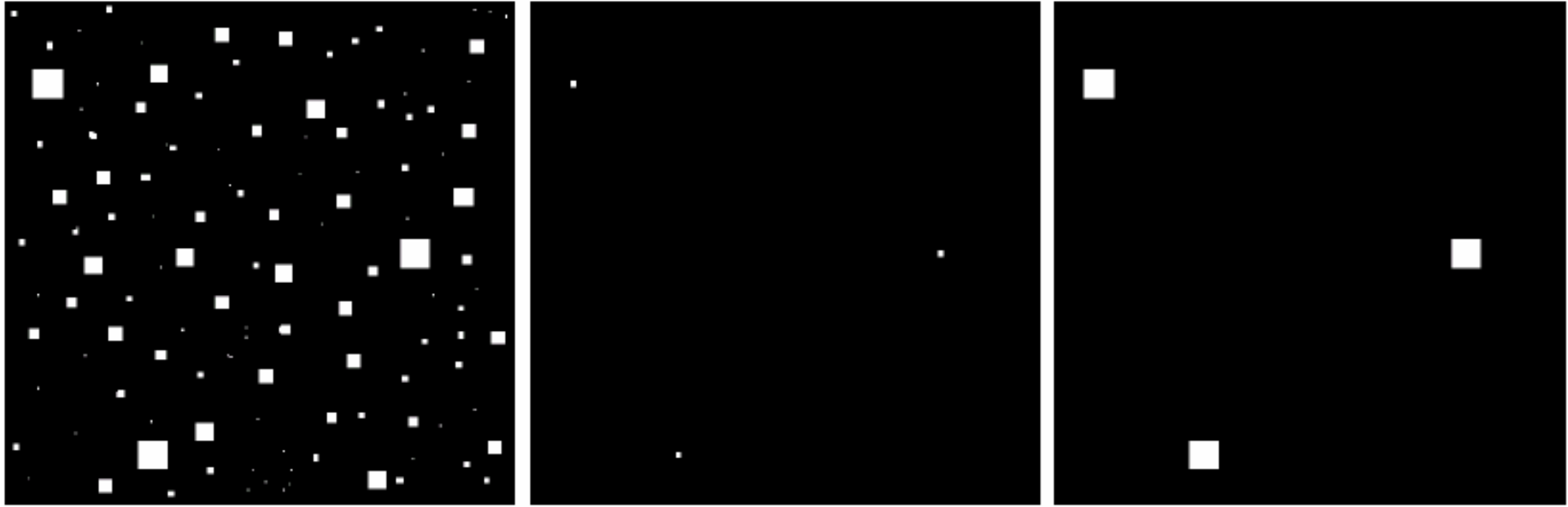
(a) Original image.

(b) Erosion with a disk of radius 10.

(c) Erosion with a disk of radius 5.

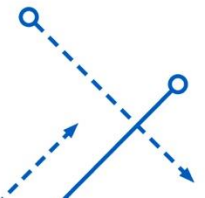
(d) Erosion with a disk of radius 20

Erosion then Dilation



a b c

- (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side.
- (b) Erosion of (a) with a square structuring element (13x13) of 1's.
- (c) Dilation of (b) with the same structuring element.



Morphology - Opening

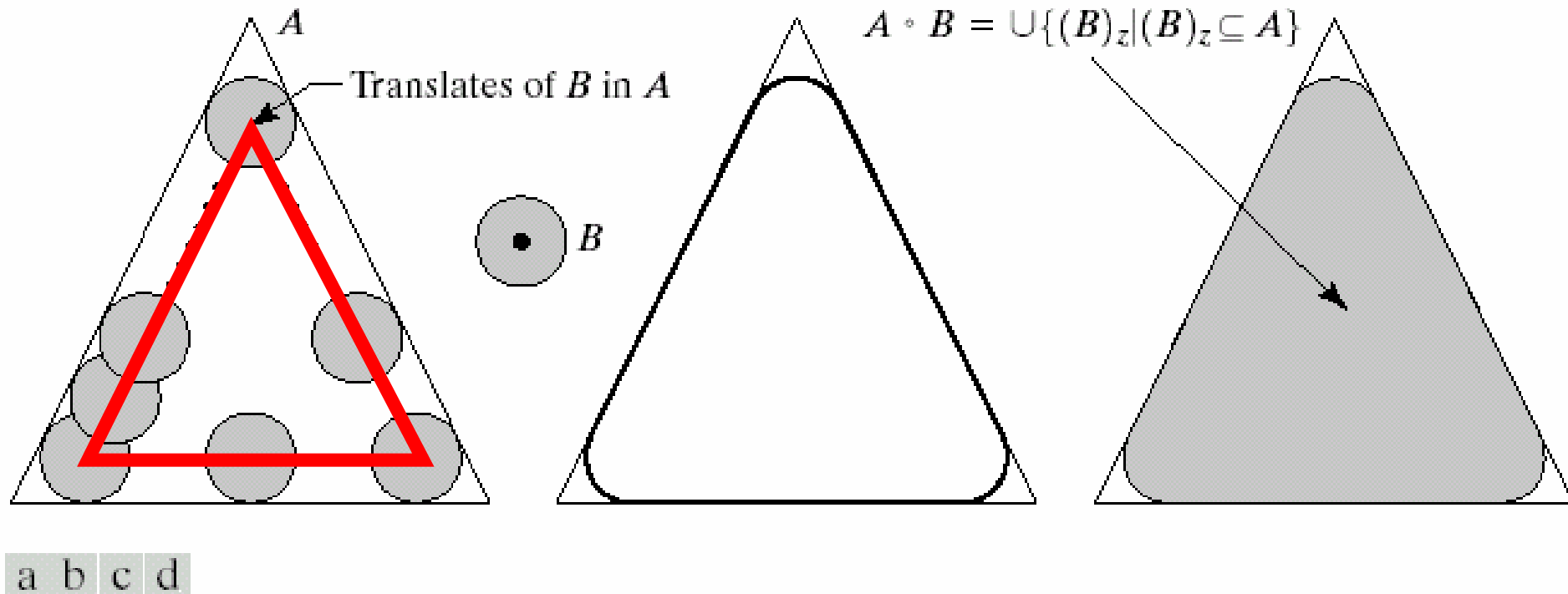
- Compound operations – Opening
- A **compound** operation is when **two or more** morphological operations are performed in **succession**.
- A common example is **opening** which is an **erosion followed by a dilation**:

$$A \circ B = (A \ominus B) \oplus B$$

- The opening A by B is obtained by taking the union of all translates of B that fit into A . This can be expressed as a fitting processing such that:

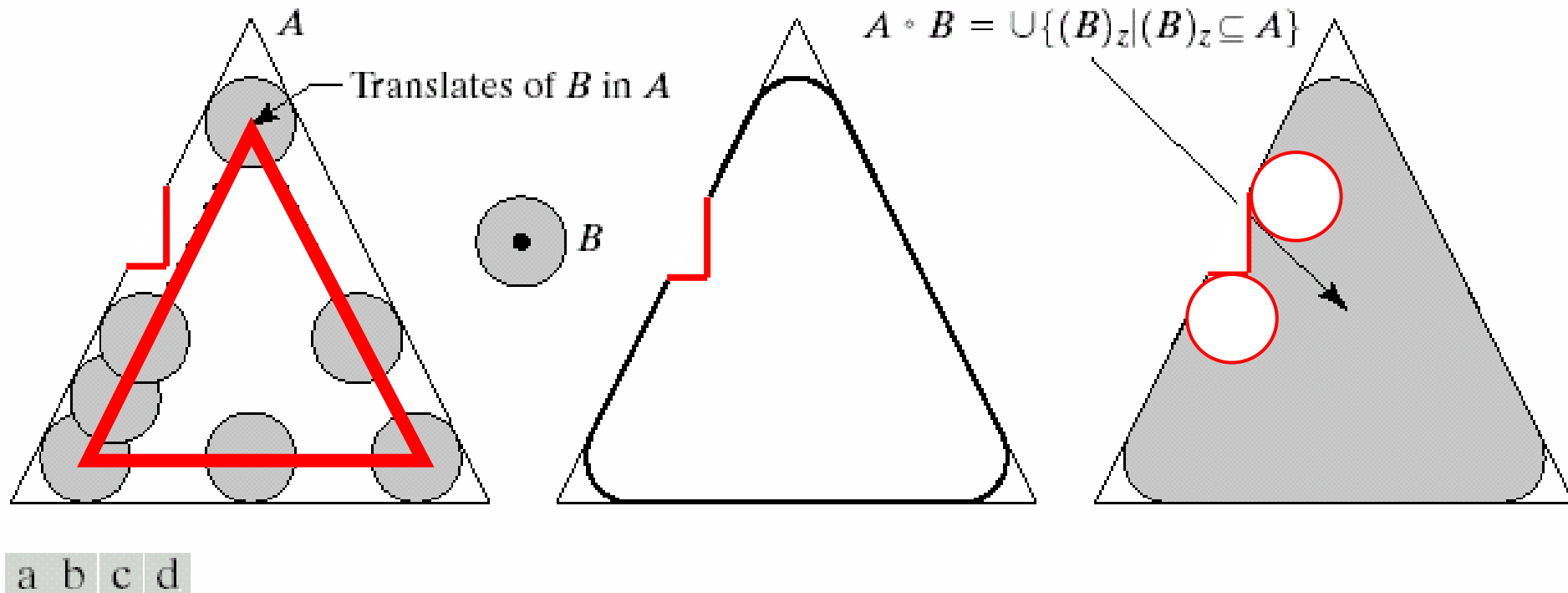
$$A \circ B = \bigcup \{ (B)_z \mid (B)_z \subseteq A \}$$

Morphology – Opening



- (ab) Structuring element B "rolling" along the inner boundary of A (the dot is the origin of B).
- (c) The heavy line is the outer boundary of the opening.
- (d) Complete opening (shaded).

Morphology – Opening



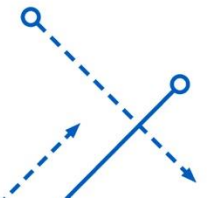
- (ab) Structuring element B "rolling" along the inner boundary of A (the dot is the origin of B).
- (c) The heavy line is the outer boundary of the opening.
- (d) Complete opening (shaded).

- Note that the outward pointing corners are rounded
- While the inward pointing corners remain unchanged.

Morphology – Opening

$$A \circ B = (A \ominus B) \oplus B \quad A \circ B = \bigcup \{(B)_z \mid (B)_z \subseteq A\}$$

- Opening is often performed to clear an image of noise whilst retaining the original object size.
- The opening operation tends to flatten the sharp peninsular projections on the object.
- Care must be taken that the operation does not distort the shape size of the object if this is significant.
- A useful way to see the effects is to look for differences between before and after opening by projecting these differences onto the original image.



Morphology – Closing

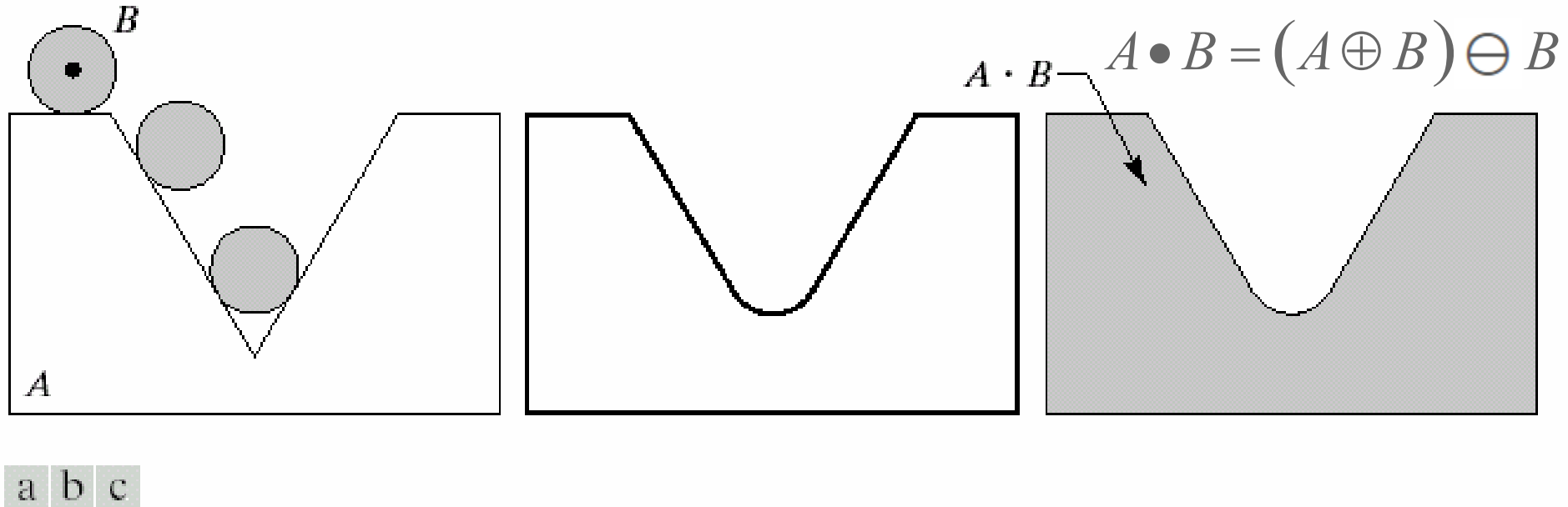
- Compound operations – Closing
- Closing is the complementary operation of opening, defined as dilation followed by erosion.

$$A \bullet B = (A \oplus B) \ominus B$$

- Opening and closing are duals of each other as:

$$(A \bullet B)^c = A^c \circ \hat{B}$$

Morphology – Closing

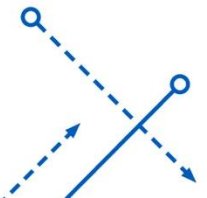


- (a) Structuring element B "rolling" on the outer boundary of set A .
- (b) Heavy line is the outer boundary of the closing.
- (c) Complete closing (shaded).

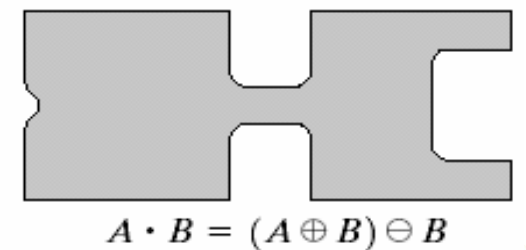
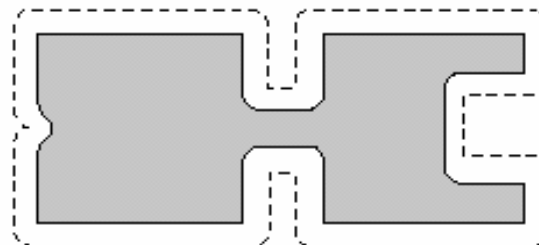
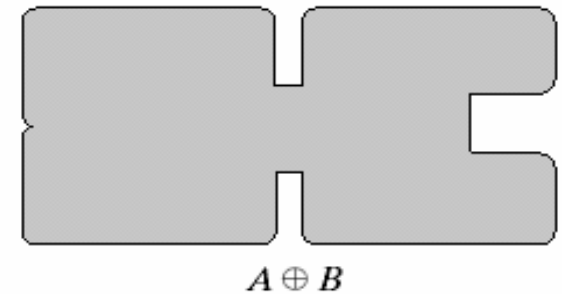
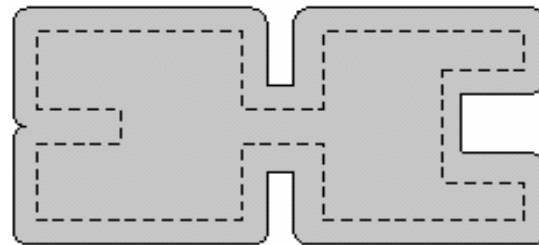
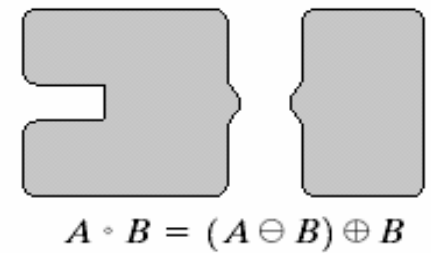
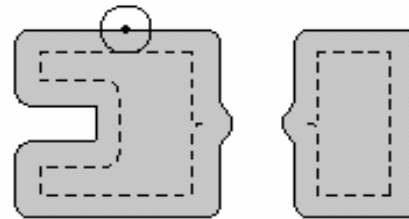
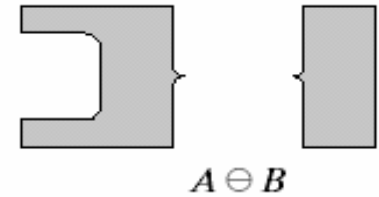
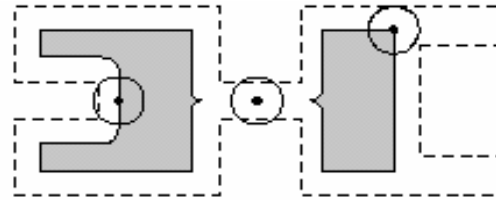
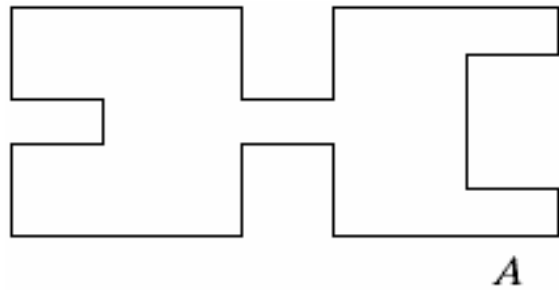
- Note that the inward pointing corners are rounded, while the outward pointing corners remain unchanged.

Morphology – Closing

- The classic application of closing is to **fill holes** in a region **whilst retaining the original object size**.
- Dilation fills the holes and erosion restores the original region size.
- In addition to filling holes the closing operation tends to fill the ‘bays (凹)’ on the edge of a region.



Erosion, Opening, Dilation, Closing



Morphology – Opening and Closing

- The opening operation satisfies the **properties**:
 - $A \circ B$ is a subset (subimage) of A
 - If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$
 - $(A \circ B) \circ B = A \circ B$
- The closing operation satisfies the **properties**:
 - A is a subset (sub image) of $A \bullet B$
 - If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$.
 - $(A \bullet B) \bullet B = A \bullet B$

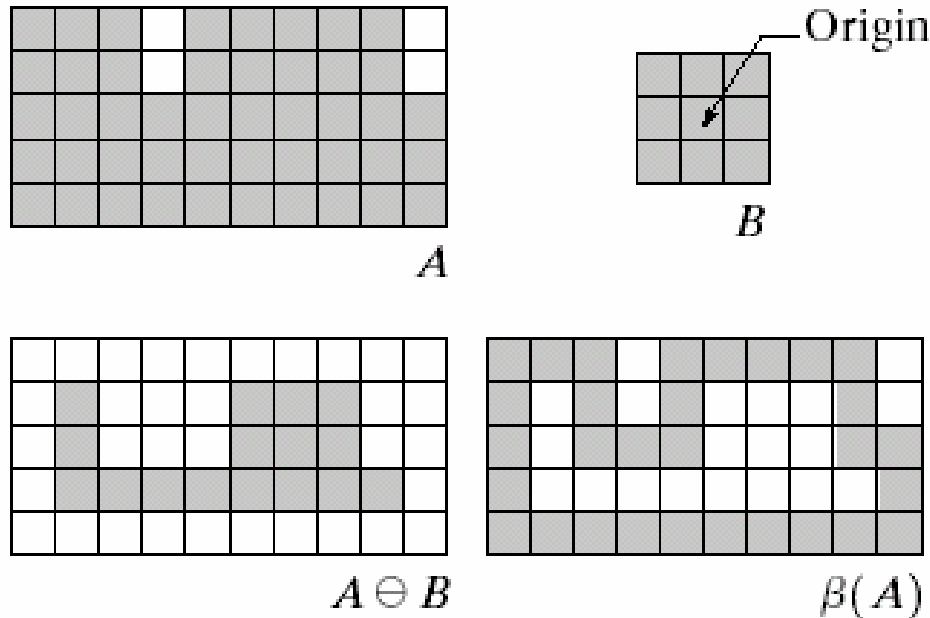
Algorithms and Applications

- Morphology can be used for many applications in image processing, pattern recognition, computer vision.
 - Boundary Extraction
 - Region Filling
 - Connected Components Extraction
 - Denoising

Boundary Extraction

- The boundary of a set A , denoted by $\beta(A)$:

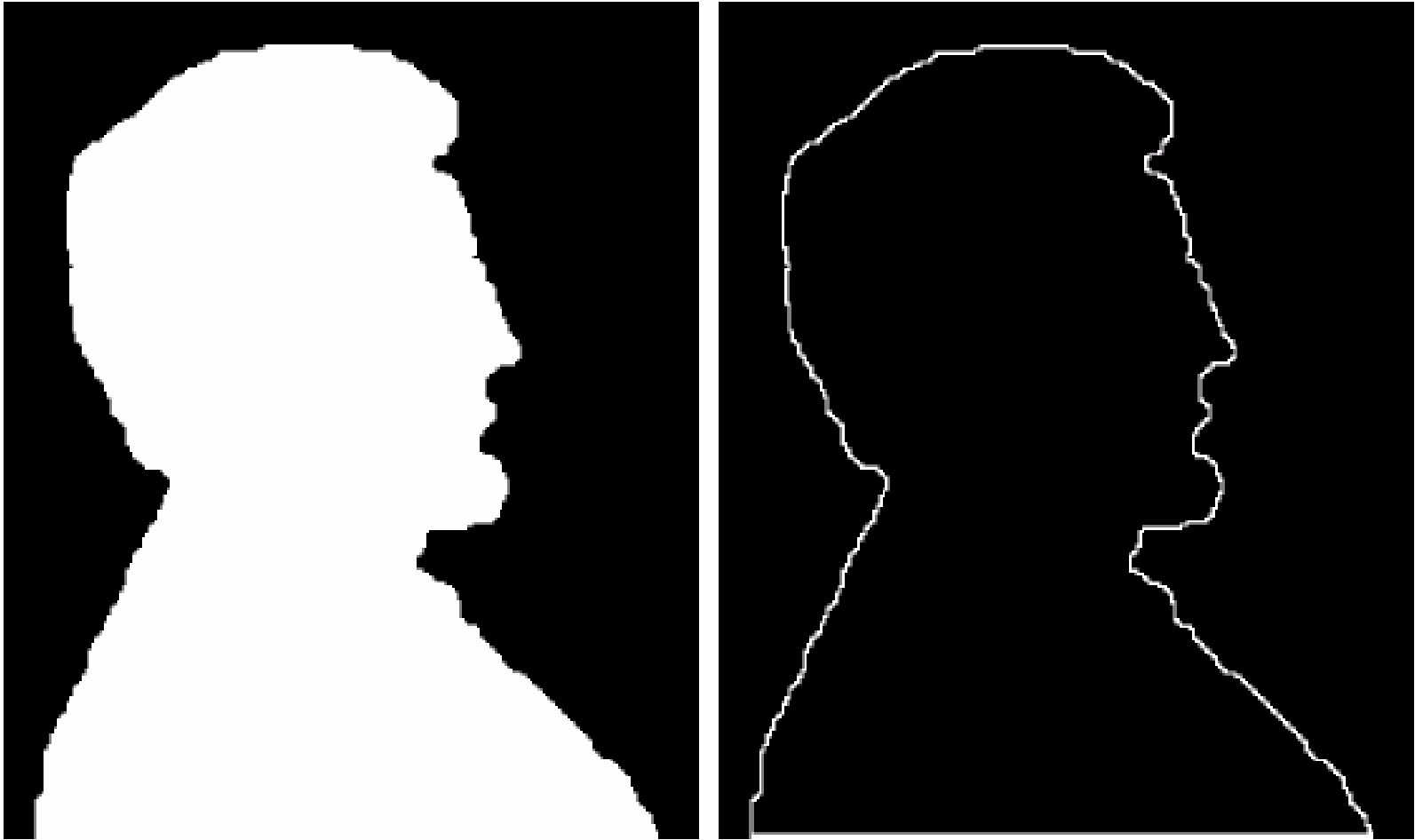
$$\beta(A) = A - (A \ominus B)$$



a	b
c	d

- Set A.
- Structuring element B.
- A eroded by B.
- Boundary, given by the set difference between A and its erosion.

Boundary Extraction - Example

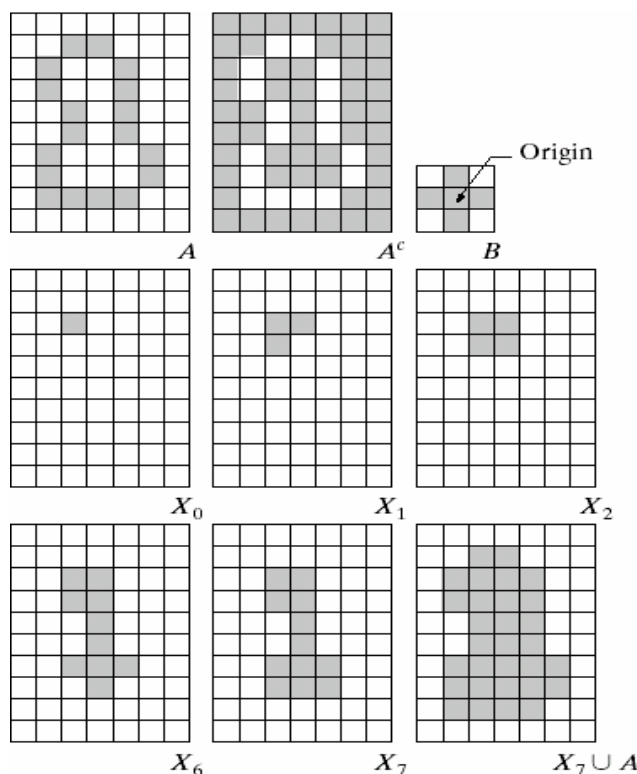


Region Filling

- Beginning with a point X_0 inside the boundary, the entire region inside the boundary is filled by the procedure:

$$A^F = X_k \cup A,$$

$$X_k = (X_{k-1} \oplus B) \cap A^c, \quad k = 1, 2, 3 \dots$$

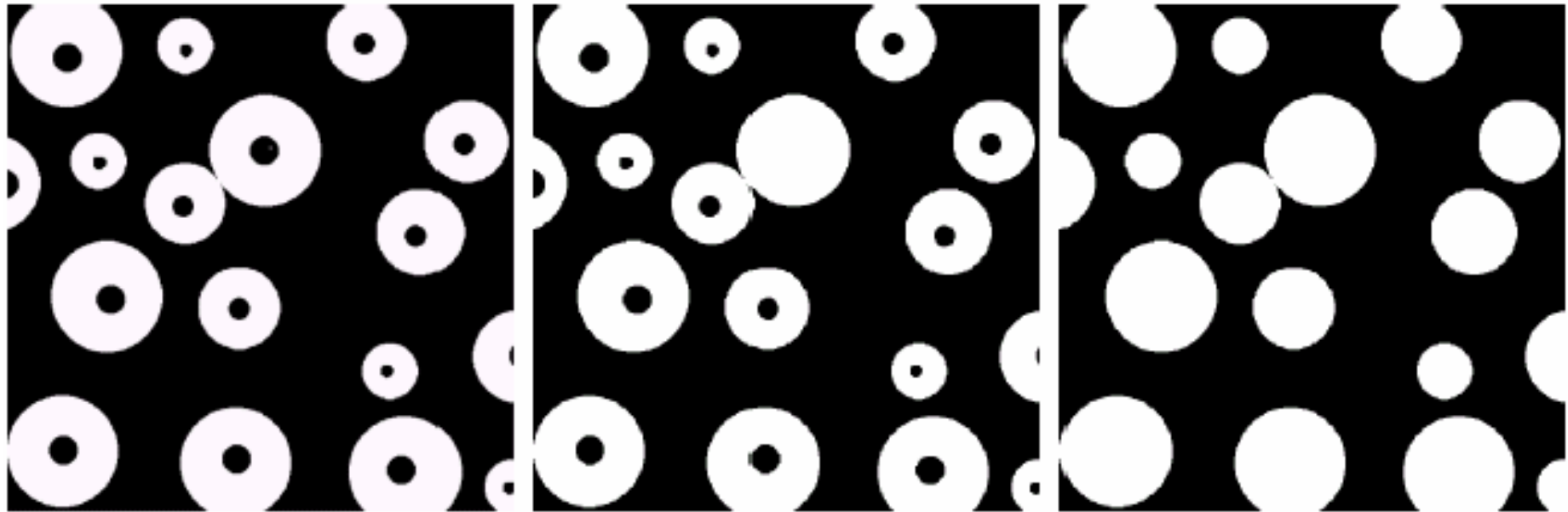


a	b	c
d	e	f
g	h	i

Region filling.

- Set A.
- Complement of A.
- Structuring element B.
- Initial point inside the boundary.
- h) Various steps of X_k .
- Final result [union of (a) and (h)]

Region Filling - Example

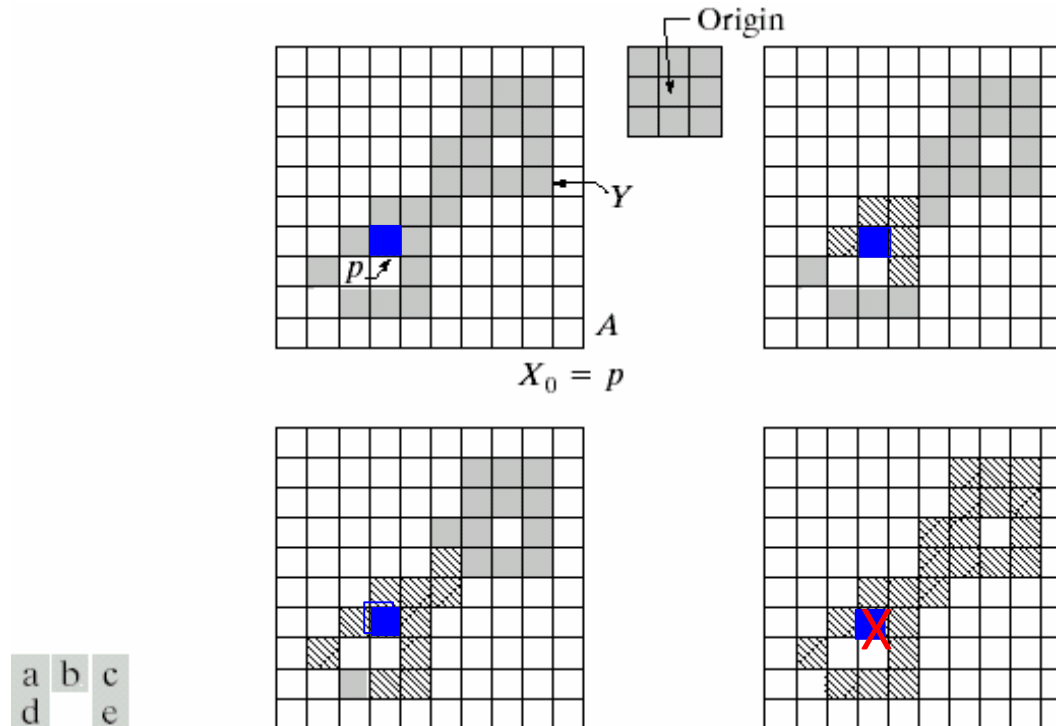


a b c

- a) Binary image (the white dot inside one of the regions is the starting point for the region-filling algorithm).
- b) Result of filling that region
- c) Result of filling all regions.

Extract connected components

- $X_k = (X_{k-1} \oplus B) \cap A, k = 1, 2, 3, \dots$



- Set A showing initial point p (all shaded points are valued 1, but are shown different from p to indicate that they have not yet been found by the algorithm).
- Structuring element.
- Result of first iterative step.
- Result of second step.
- Final Results.

Denoising

- Closing and Opening can be used to **eliminate noise**.

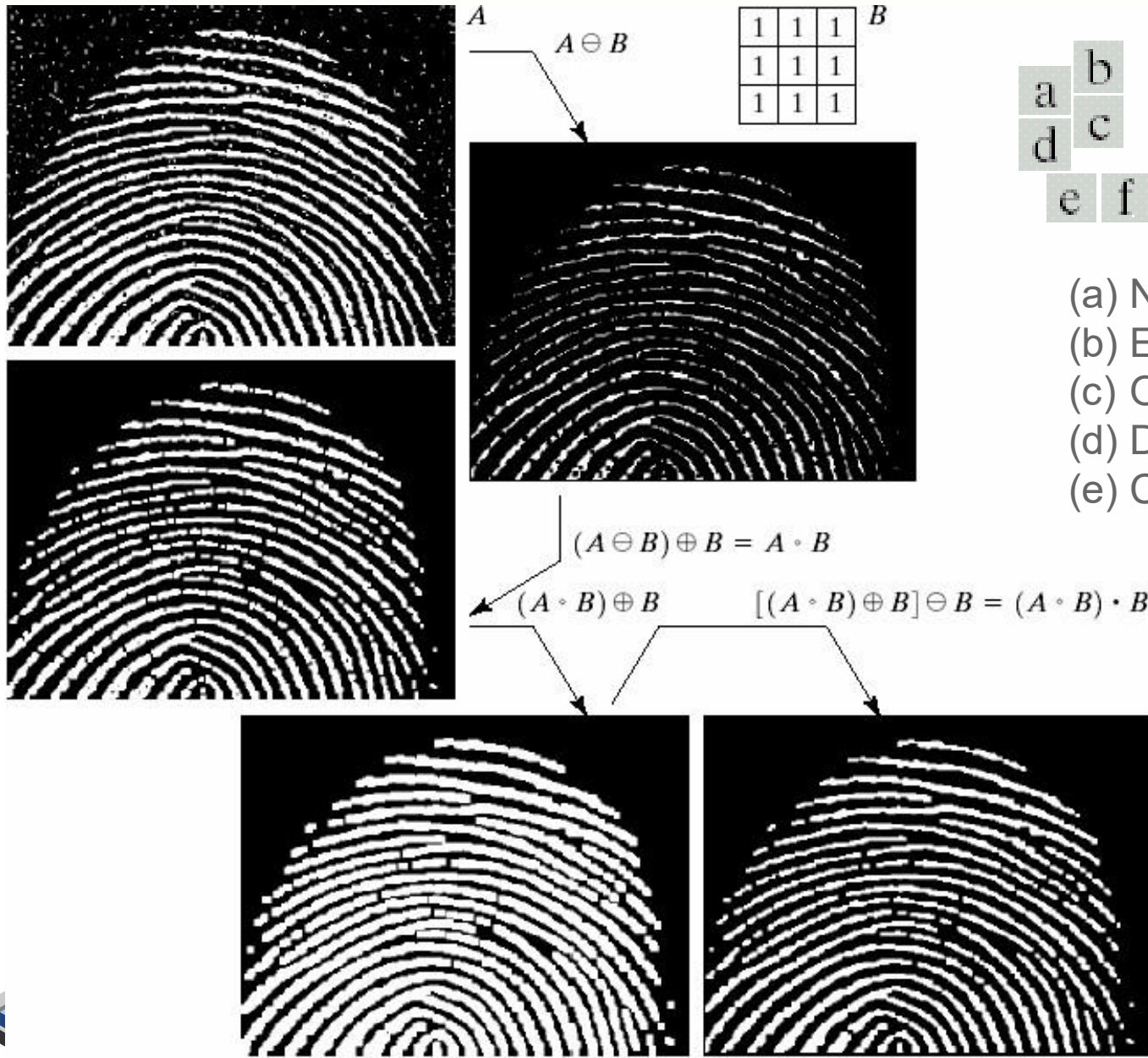
$$(A \circ B) \bullet B$$

or

$$(A \bullet B) \circ B$$

- Noise **outside** the object are removed by **opening** with B
- Noise **inside** the object are removed by **closing** with B .

Algorithms and Applications



- (a) Noisy image.
- (b) Eroded image.
- (c) Opening of A .
- (d) Dilation of the opening.
- (e) Closing of the opening.

Morphology – Summary

		Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Operation	Equation	
Translation	$(A)_z = \{w \mid w = a + z, \text{ for } a \in A\}$	Translates the origin of A to point z .
Reflection	$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement	$A^c = \{w \mid w \notin A\}$	Set of points not in A .
Difference	$A - B = \{w \mid w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation	$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$	“Expands” the boundary of A . (I)
Erosion	$A \ominus B = \{z \mid (B)_z \subseteq A\}$	“Contracts” the boundary of A . (I)

Morphology – Summary

Opening	$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing	$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Region filling	$X_k = (X_{k-1} \oplus B) \cap A^c; X_0 = p$ and $k = 1, 2, 3, \dots$	Fills a region in A , given a point p in the region. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A; X_0 = p$ and $k = 1, 2, 3, \dots$	Finds a connected component Y in A , given a point p in Y . (I)